

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Requirement Analysis

Solution Requirements

Project Name	Visualizing Housing Market Trends
Team ID	
Date	14 July 2026

Functional Requirement

#	Requirement	Details
1.	Data Collection Interface	- Import residential property sales dataset from cloud storage (Google Drive). - Connect the data source directly to Tableau for live refresh.
2.	Data Storage & Management	- Centralized cloud-based storage for raw housing sales data. - Capability to update records as new listings/sales are added.
3.	Data Cleaning & Preprocessing	- Remove duplicates and null values. - Standardize fields such as renovation year, floors, and bathroom/bedroom counts.
4.	Interactive Visualization (Tableau)	- Dashboards showing sale price trends, house age distribution, and renovation impact. - Filters by house age, renovation status, and structural features. - Trend analysis and storyboards.
5.	Analytics & Insights	- Pattern recognition (e.g., price bands most affected by renovation). - Comparison of renovated vs. non-renovated homes across age groups.

Non-Functional Requirement

#	Requirement	Details
1.	Scalability	Should handle sales records for hundreds or thousands of properties at a time.
2.	User-Friendliness	Dashboard and reports should be easy to navigate, with minimal training required.
3.	Performance	Fast data processing and dashboard loading, even for large datasets.
4.	Data Privacy & Security	Secure login and role-based access control to sales and property data.

5.	Compatibility	Should work on various devices (laptops, tablets, mobiles) and support data export (PDF, Excel).
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